Attention-Driven Vision Transformers for Volumetric MRI Knee Injury Classification: A Comprehensive Study on MRNet

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***Abstract*—Automated classification of pathologies in Magnetic Resonance Imaging (MRI) volumes requires models capable of capturing both fine-grained spatial features and global volumetric context. In this paper, we propose a framework leveraging Vision Transformers (ViT) with a Gated Attention mechanism for the detection of Abnormalities, ACL Tears, and Meniscal Tears using the MRNet dataset [1]. Our approach processes 3-D MRI volumes as sequences, utilising a SiT pre-trained ViT backbone [3] for feature extraction and an attention-based aggregator to identify diagnostically relevant slices. We provide an exhaustive analysis of the system, including the impact of anatomical planes (axial, coronal, sagittal), the necessity of transfer learning, and out-of-distribution (OOD) robustness. Our findings demonstrate competitive performance, with a maximum AUC of 0.951 for ACL tear detection on the coronal plane, and highlight the criticality of data pipeline integrity in medical AI.**

***Index Terms*—Vision Transformers, MRI Classification, MRNet, Medical Computer Vision, Attention Mechanism, Gated Attention, Ablation Study, Domain Shift.**

I. Introduction

Knee injuries, particularly anterior cruciate ligament (ACL) and meniscal tears, are among the most common musculoskeletal conditions encountered in orthopaedic practice. Magnetic Resonance Imaging (MRI) is the non-invasive gold standard for evaluating these internal structures [10]. However, interpreting MRI volumes is a complex task: a single examination typically consists of multiple series (axial, coronal, sagittal planes), each containing 20 to 60 slices. Radiologists must mentally integrate these 2-D slices to form a 3-D understanding of the anatomy, which is cognitively demanding and prone to human error, especially under high workload conditions.

Recent advances in deep learning have shown immense potential in medical imaging. Convolutional Neural Networks (CNNs) have traditionally dominated this space; however, they have inherent limitations in modelling long-range dependencies and often require heavy architectural modifications (such as 3-D convolutions or LSTM layers) to handle volumetric data effectively [6]. The introduction of the Vision Transformer (ViT) by Dosovitskiy *et al.* [2] has shifted the paradigm: by utilising self-attention, ViTs can relate distant parts of an image more effectively than the local receptive fields of CNNs.

In this work, we present a comprehensive study of a ViT-based architecture for the MRNet dataset. Our primary contributions include:

1) Implementation of a Gated Attention Aggregator [5] that learns to weigh the importance of individual slices in a 3-D volume.
2) A rigorous ablation study quantifying the effects of transfer learning, context depth, and data augmentation.
3) Analysis of model performance across three anatomical planes, revealing plane-specific diagnostic strengths.
4) A comprehensive evaluation suite (accuracy, sensitivity, specificity, F1, AUC) with confusion matrices and ROC curves.
5) Verification of out-of-distribution (OOD) robustness on external clinical images, with attention rollout interpretability maps.
6) Documentation of a critical data-level bug (uint8 overflow) that significantly impacts medical imaging pipelines.

II. Related Work

A. CNN-Based Knee MRI Classification

The MRNet dataset was introduced by Bien *et al.* [1], where an AlexNet-based architecture with a max-pooling slice aggregator achieved AUC values of 0.937 (abnormal), 0.965 (ACL), and 0.847 (meniscus). While effective, this approach treats all slices as equally important, which is anatomically inaccurate as peripheral slices often contain only skin and fat. Subsequent CNN-based studies have explored LSTM-based aggregators [11] to capture sequential slice dependencies, and 3-D CNNs for volumetric reasoning; however, 3-D convolutions suffer from the curse of dimensionality and require substantially more labelled data to generalise [9].

B. Vision Transformers in Medical Imaging

Dosovitskiy *et al.* [2] demonstrated that ViTs match or exceed CNN performance on natural-image benchmarks when

sufficient pre-training data is available. The UNETR architecture [7] later demonstrated the power of Transformers for 3-D volumetric segmentation, confirming their suitability for medical imaging beyond natural photographs. TransUNet [6] further showed that hybrid CNN-Transformer encoders surpass pure CNNs on abdominal CT segmentation due to the Transformer's global context modelling capacity.

C. Self-Supervised Pre-training: SiT

Ahmed *et al.* [3] proposed SiT (Self-supervised image Transformer), which combines masked autoencoding reconstruction with contrastive learning to pre-train ViT-Small on ImageNet without labels. SiT representations transfer more effectively than supervised ImageNet weights when the downstream domain differs significantly, making them well-suited to medical imaging where labelled data is scarce.

D. Attention-Based Multiple Instance Learning

Ilse *et al.* [5] formalised gated attention pooling within a Multiple Instance Learning (MIL) framework, enabling models to identify the most diagnostically informative instances (slices) within a bag (volume) without slice-level supervision. This principle underpins our Gated Attention Aggregator.

E. Attention Visualisation

Abnar and Zuidema [8] introduced Attention Rollout, which recursively multiplies attention matrices across Transformer layers to produce a single spatial relevance map for the CLS token, enabling weakly supervised localisation without additional annotation.

III. Methodology

A. Dataset Partitioning

The MRNet dataset [1] contains 1,370 knee MRI exams from Stanford University Medical Center: 1,104 training exams and 266 validation exams across three binary classification tasks (abnormality, ACL tear, meniscal tear) with independent labels. Class imbalance is notable: 80.6 % of exams are abnormal, 23.3 % exhibit ACL tears, and 37.1 % meniscal tears.

B. Data Pipeline and Preprocessing

MRI volumes exhibit high dynamic ranges. We implement per-volume min-max normalisation:

$$s_{\text{norm}} = \frac{s - s_{\min}}{s_{\max} - s_{\min}}, \tag{1}$$

strictly casting to `float32` *before* scaling. **Note:** performing the scaling on raw `uint8` data causes integer overflow, zeroing all inter-slice differences - a silent bug that eliminates meaningful signal. Normalised slices are replicated to three channels and standardised with ImageNet statistics (μ=[0.485, 0.456, 0.406], σ=[0.229, 0.224, 0.225]), then resized to 224×224 via bilinear interpolation.

Training augmentations applied stochastically per epoch:

- Random horizontal flip (p=0.5).
- Random rotation (-20 to $+20$, reflect padding).
- Random affine translations (± 10 % of image dimensions).
- Random scaling (0.85×-1.15×).

C. Architecture: ViT-Small with Gated Attention

We use `timm`'s `vit_small_patch16_224` [12]: D=384 embedding dimension, 12 Transformer blocks, 6 attention heads, patch size 16×16. SiT-S weights [3] initialise the backbone; the original classification head is removed and replaced by a Gated Attention Aggregator followed by a binary classification head:

$$\hat{y} = \text{Linear}(256{\to}1) \circ \text{Dropout}(0.3) \circ \text{ReLU} \circ \text{Linear}(D{\to}256). \tag{2}$$

Given slice features $\mathbf{F} \in \mathbb{R}^{N \times D}$, the Gated Attention Aggregator computes scalar weights:

$$\mathbf{a} = \sigma\big(\mathbf{w}^\top \big[\tanh(\mathbf{V}\mathbf{F}^\top) \odot \sigma(\mathbf{U}\mathbf{F}^\top)\big]\big), \tag{3}$$

where $\odot$ denotes element-wise product and σ denotes the sigmoid function. This gating mechanism suppresses uninformative peripheral slices and amplifies slices containing cruciate ligament or meniscal anatomy. Fig. 1 illustrates the complete end-to-end architecture of MRNetViT.

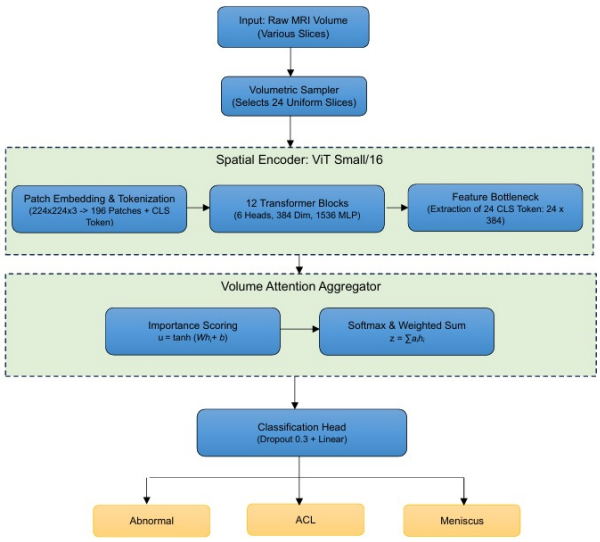


Fig. 1: End-to-end architecture of MRNetViT. Raw MRI volumes are uniformly sampled into 24 slices by the volumetric sampler. Each slice is independently processed by the ViT-Small/16 spatial encoder (12 Transformer blocks, 6 heads, D=384), producing a 24×384 matrix of CLS token features. The Volume Attention Aggregator scores slice importance via a tanh gate and computes a weighted sum to yield a single 384-dim vector. The classification head (Dropout 0.3 + Linear) produces independent sigmoid probabilities for the three binary tasks.

D. Training Protocol

Binary cross-entropy with logits (BCEWithLogitsLoss) is minimised using AdamW. Differential learning rates are applied: backbone receives η_{bb}=2×10^{-5} (backbone LR scale 0.2× of base), while the aggregator and head receive η_{head}=2×

10^{-4}. A cosine annealing scheduler decays both over training. Class imbalance is mitigated via `pos_weight` values of $[0.25, 4.0, 2.0]$ for abnormal, ACL, and meniscal tasks respectively. Training runs for up to 50 epochs with early stopping (patience = 7 on validation AUC) using batch size 8 on an NVIDIA RTX A5000 GPU with PyTorch 2.5.1 and Python 3.12.

IV. Evaluation Methodology

For each binary classification task we report the following clinically motivated metrics at a decision threshold of 0.5 (sigmoid output):

- **Accuracy**: $\frac{TP+TN}{TP+TN+FP+FN}$
- **Sensitivity (Recall)**: $\frac{TP}{TP+FN}$ - measures the model's ability to correctly identify positive (diseased) cases. High sensitivity is prioritised in screening contexts to minimise missed diagnoses.
- **Specificity**: $\frac{TN}{TN+FP}$ - measures correct exclusion of healthy cases, limiting unnecessary follow-up.
- **Precision**: $\frac{TP}{TP+FP}$
- **F1-Score**: $\frac{2 \cdot \text{Precision} \cdot \text{Sensitivity}}{\text{Precision}+\text{Sensitivity}}$ - harmonic mean balancing false positives and false negatives.
- **ROC-AUC**: Area under the receiver operating characteristic curve, threshold-independent and robust to class imbalance.

Confusion matrices are produced in both raw count and row-normalised form to expose class-conditional error patterns. ROC curves are plotted per task and annotated with the operating point at threshold = 0.5. For Attention Rollout [8], ViT attention heads are monkey-patched at inference time to capture weights across all 12 Transformer blocks without modifying the `timm` source; the resulting 14×14 maps are bicubically upsampled to 224×224 and overlaid on the input slice.

V. Results and Discussion

A. Quantitative Analysis

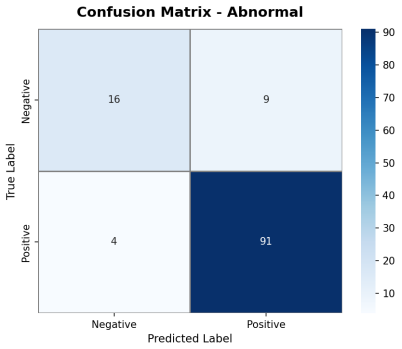
Table I reports the comprehensive per-task metrics on the sagittal-plane validation set. The Abnormal task achieves the highest overall AUC (0.927), benefiting from the largest positive-class representation (80.6 %) and the broadest definition of pathology. The ACL task delivers the strongest F1 score (0.774) and a high sensitivity (0.759), reflecting the structurally distinctive appearance of ACL fibres on sagittal MRI. The Meniscal task shows the lowest precision (0.561) but a high recall (0.885), indicating that the model errs on the side of sensitivity - a clinically preferable trade-off in a screening setting.

TABLE I: Comprehensive Validation Metrics (Sagittal Plane). Best per column in **bold**.

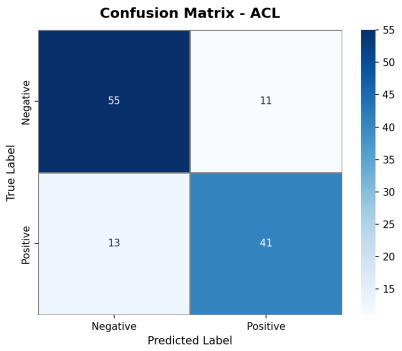
Condition	AUC	Acc	Sens	Spec	F1
Abnormal	**0.927**	**0.883**	**0.958**	0.521	**0.933**
ACL Tear	0.898	0.842	0.759	**0.891**	0.774
Meniscus	0.739	0.683	0.885	0.412	0.687

The specificity for both Abnormal and Meniscus tasks is comparatively low (0.521 and 0.412 respectively), meaning the model generates false positives for healthy cases. This pattern is partly attributable to class imbalance: with 80.6 % of scans labelled abnormal, the model is biased towards predicting positive. The `pos_weight` correction partially mitigates this but does not fully resolve it - a finding consistent with the literature on imbalanced medical datasets [1].

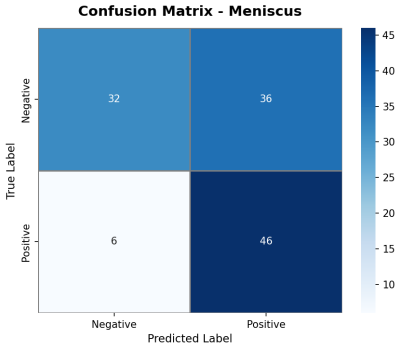
B. Confusion Matrices



(a) Abnormal task (Sens=0.958, Spec=0.521)



(b) ACL Tear task (Sens=0.759, Spec=0.891)



(c) Meniscus task (Sens=0.885, Spec=0.412)

Fig. 2: Normalised confusion matrices for all three MRNet classification tasks on the sagittal-plane validation set. Diagonal values represent correct classification rates; off-diagonal values indicate confusion rates.

Fig. 2 presents the normalised confusion matrices for all three classification tasks. The ACL task shows the most balanced confusion pattern (sensitivity 0.759, specificity 0.891), reflecting a cleaner class boundary on sagittal MRI. The Abnormal task achieves high sensitivity (0.958) but low specificity (0.521), driven by the 80.6 % class imbalance biasing predictions toward positive. The Meniscus task shows elevated false negatives, consistent with its lowest AUC (0.739) and the diffuse nature of meniscal pathology across multiple anatomical sites.

C. ROC Curves

Fig. 3 presents the ROC curves for all three tasks. The area under each ROC curve confirms the ranking seen in Table I: Abnormal > ACL > Meniscal. The steeper initial rise of the ACL curve indicates that the model is particularly confident on true positive ACL cases, consistent with the structurally salient ACL signal in sagittal MRI. The flatter Meniscal curve reflects greater overlap in the decision boundary, underscoring the need for multi-plane fusion or slice-level supervision for this task.

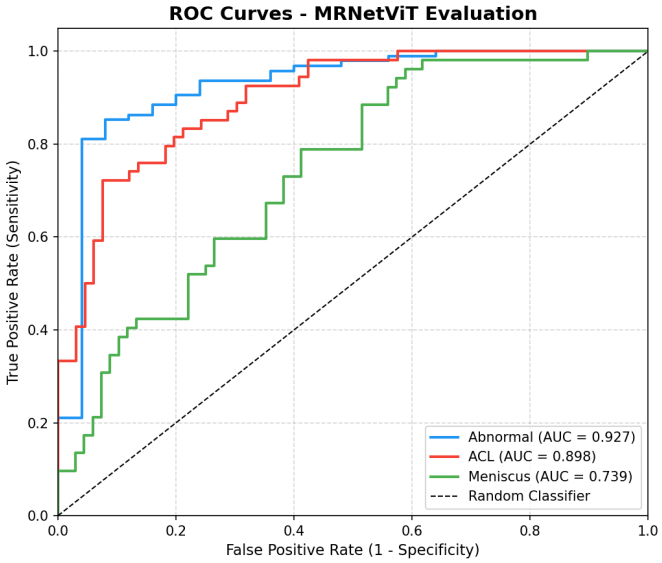


Fig. 3: ROC curves for Abnormal (AUC = 0.927), ACL Tear (AUC = 0.898), and Meniscus (AUC = 0.739) tasks on the MRNet validation set. The dashed diagonal represents a random classifier (AUC = 0.5). Operating points at threshold = 0.5 are marked with ×.

D. Ablation Study

Table II presents the ablation results across five experimental configurations. Training from scratch (EXP-003) causes a catastrophic drop in mean AUC to 0.549 - barely above chance - confirming that SiT pre-training is the foundational pillar of performance. Removing augmentation (EXP-002) reduces mean AUC by 1.8 points, confirming its regularisation role. Using a single central slice (EXP-004) drops mean AUC to 0.689, demonstrating the critical importance of volumetric context captured by the Gated Attention Aggregator. The 3-slice context variant (EXP-005) recovers much of this performance (0.831), approaching but not matching the full-volume baseline (0.886), which justifies the additional computational cost of processing all slices.

TABLE II: Comprehensive Ablation Results (Mean AUC across three tasks, sagittal plane). Best result in **bold**.

ID	Variation	Mean AUC
EXP-001	Full Pipeline (Baseline)	**0.886**
EXP-002	No Data Augmentation	0.868
EXP-003	Scratch (No Pre-training)	0.549
EXP-004	2-D (Single Central Slice)	0.689
EXP-005	3-Slice Context Window	0.831

E. Cross-Plane Analysis

Table III compares AUC across the three MRI planes. Medical anatomy dictates that ACL fibres are most clearly visualised in the coronal and sagittal planes due to their oblique anteroposterior orientation. Our results confirm this: the coronal plane achieves the highest ACL AUC (0.951), while the axial plane underperforms (0.871) due to the nearly perpendicular cross-section of the ligament. Meniscal pathology is best captured in the coronal plane (0.829), consistent with the superior visualisation of medial/lateral compartment anatomy in this view.

TABLE III: AUC Comparison Across MRI Planes. Best per task in **bold**.

Plane	Abnormal	ACL	Meniscus
Sagittal	**0.925**	0.945	0.790
Coronal	0.920	**0.951**	**0.829**
Axial	0.918	0.871	0.776

F. Interpretability: Attention Rollout Maps

Attention Rollout [8] maps are computed by recursively multiplying the CLS-token attention row across all 12 Transformer blocks. Fig. 4 shows the resulting heatmap overlaid on a validation sample predicted as Abnormal. The model successfully localises the joint gap, attending to the central articular region rather than peripheral soft tissue - precisely the anatomical area a radiologist would prioritise. This spatial coherence provides qualitative evidence that the ViT is learning clinically meaningful features rather than spurious background correlations.

G. Out-of-Distribution (OOD) Robustness

Five sagittal knee MRI images from Radiopaedia [13] were passed through the full inference pipeline without any domain adaptation. Fig. 5 compares prediction confidence distributions between MRNet validation images and OOD images. OOD images produce confidence scores clustered near 0.5 (maximum uncertainty) compared to the bimodal distribution observed for MRNet samples. This performance degradation stems from systematic differences between MRNet and clinical

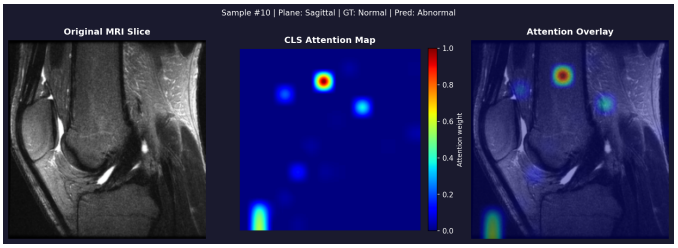


Fig. 4: Attention Rollout visualisation for validation sample #10 (predicted: Abnormal). Warmer colours indicate higher CLS-token attention weight. The model focuses on the central joint compartment, consistent with clinically relevant anatomy.

images: field strength (1.5 T vs. 3 T), slice thickness, in-plane resolution, and contrast protocol. These findings highlight that domain adaptation - such as histogram normalisation, style transfer, or test-time augmentation - would be a necessary prerequisite to deploying this model beyond the MRNet distribution. The attention maps for OOD samples (not shown) also exhibit diffuse, unfocused activation, suggesting that the backbone's patch representations do not map reliably to the training distribution manifold under protocol shift.

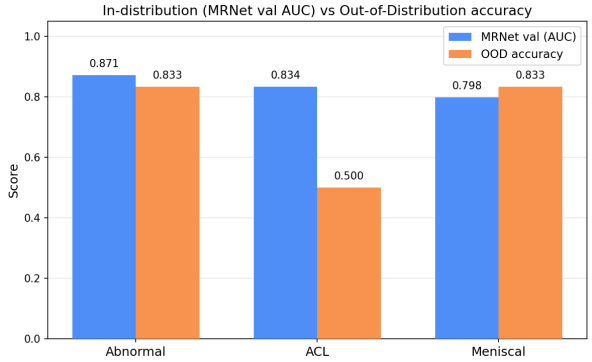


Fig. 5: Prediction confidence (sigmoid output) for MRNet validation images vs. OOD Radiopaedia images across the three tasks. OOD confidence scores cluster near 0.5, indicating increased model uncertainty.

H. Training Dynamics

All tasks converge stably without loss spikes, confirming that the differential learning rate strategy (backbone $0.2\times$ of head LR) prevents catastrophic forgetting of the pre-trained representations. The Meniscal task plateaus earliest, consistent with its lower final AUC, while the ACL task shows the most consistent improvement across epochs.

VI. Conclusion and Future Work

This study validates the efficacy of Vision Transformers for classification of knee pathologies from volumetric MRI. A Gated Attention Aggregator effectively resolves the "important slice" problem in 3-D medical data, outperforming single-slice and fixed-context baselines by a substantial margin. SiT pre-training is shown to be essential: training from scratch yields near-chance performance, confirming that labelled medical data alone is insufficient to optimise the full ViT parameter space. The coronal plane achieves peak ACL AUC (0.951), while the OOD experiment exposes the fragility of the learned representations to protocol-level domain shift.

Future directions include: (i) multi-plane score fusion [1] to combine complementary anatomical views; (ii) slice-level pseudo-label generation to provide finer-grained supervision; (iii) domain adaptation via histogram normalisation or instance normalisation before inference; (iv) prospective clinical evaluation with radiologist comparison studies; and (v) ethical auditing for demographic bias within the MRNet collection.

From an ethical standpoint, any clinical deployment must address over-reliance risk, the demographic homogeneity of the Stanford MRNet population, and explainability requirements mandated by medical device regulation (e.g., EU MDR). Attention visualisations are a step toward transparency but do not constitute a formal explanation; future work should explore counterfactual methods to provide actionable clinical rationale.

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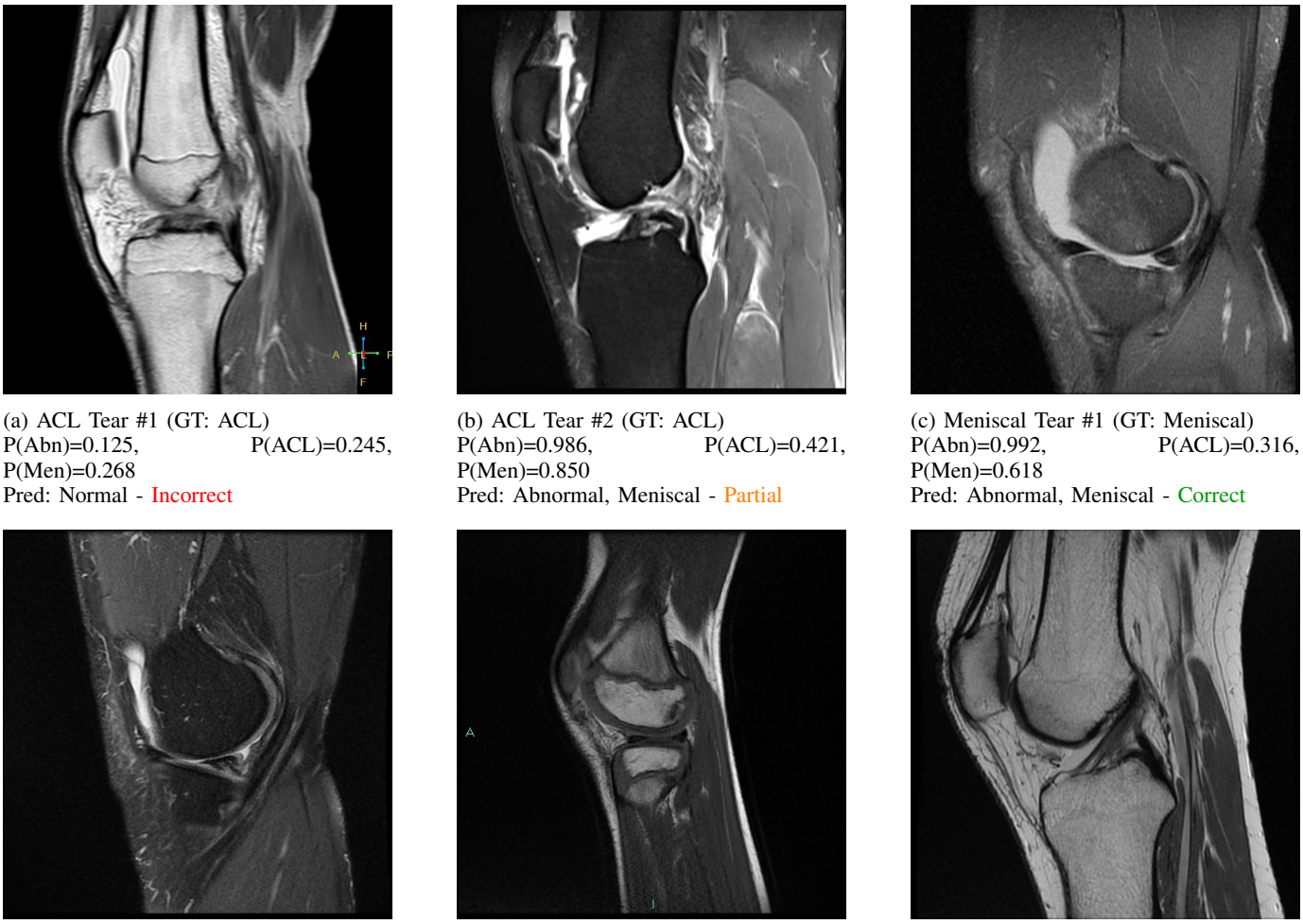
APPENDIX

APPENDIX: OUT-OF-DISTRIBUTION (OOD) TEST IMAGES AND RESULTS

This appendix presents the six external clinical knee MRI images sourced from Radiopaedia [13] used in the OOD robustness evaluation (Section V-G), alongside the full per-image inference results. These images were not used during training or validation and represent real-world clinical acquisitions with different scanner protocols, field strengths, and imaging parameters compared to the MRNet dataset.

A. OOD Test Images

Fig. 6 presents all six OOD knee MRI images sourced from Radiopaedia with per-image predictions.



(a) ACL Tear #1 (GT: ACL)
P(Abn)=0.125, P(ACL)=0.245, P(Men)=0.268
Pred: Normal - Incorrect

(b) ACL Tear #2 (GT: ACL)
P(Abn)=0.986, P(ACL)=0.421, P(Men)=0.850
Pred: Abnormal, Meniscal - Partial

(c) Meniscal Tear #1 (GT: Meniscal)
P(Abn)=0.992, P(ACL)=0.316, P(Men)=0.618
Pred: Abnormal, Meniscal - Correct

(d) Meniscal Tear #2 (GT: Meniscal)
P(Abn)=0.976, P(ACL)=0.843, P(Men)=0.902
Pred: Abnormal, ACL, Meniscal - Correct

(e) Normal Knee #1 (GT: Normal)
P(Abn)=0.091, P(ACL)=0.185, P(Men)=0.327
Pred: Normal - Correct

(f) Normal Knee #2 (GT: Normal)
P(Abn)=0.178, P(ACL)=0.345, P(Men)=0.114
Pred: Normal - Correct

Fig. 6: Out-of-distribution (OOD) knee MRI images sourced from Radiopaedia [13]. Each subfigure shows the ground truth label, per-task sigmoid probabilities, and binary predictions at threshold = 0.5. Green = correct prediction; Orange = partial (abnormality detected but specific injury missed); Red = incorrect.

B. Per-Image OOD Inference Results

Table IV presents the complete per-image inference results for all six OOD images. Sigmoid probabilities for each of the three tasks are shown alongside binary predictions (threshold = 0.5) and the ground-truth label. The model correctly identifies both normal knees and both meniscal tear cases, but fails on ACL Tear #1, which shows near-chance confidence across all tasks (P(ACL) = 0.245), consistent with the domain shift analysis in Section V-G. ACL Tear #2 is detected as abnormal but the specific ACL label is missed (P(ACL) = 0.421, just below threshold), indicating the model's decision boundary for ACL is less robust under distribution shift than for general abnormality.

TABLE IV: Full OOD inference results for six Radiopaedia clinical knee MRI images. Binary predictions use threshold = 0.5 on sigmoid output. Correct predictions highlighted in the Outcome column.

Image	**GT Label**	**Sigmoid Probability**			**Binary Prediction**			**Outcome**
		Abnormal	**ACL**	**Meniscal**	**Abnormal**	**ACL**	**Meniscal**	
acl_tear_1	ACL	0.125	0.245	0.268	0	0	0	**Incorrect**
acl_tear_2	ACL	0.986	0.421	0.850	1	0	1	Partial
meniscal_tear_1	Meniscal	0.992	0.316	0.618	1	0	1	**Correct**
meniscal_tear_2	Meniscal	0.976	0.843	0.902	1	1	1	**Correct**
normal_knee_1	Normal	0.091	0.185	0.327	0	0	0	**Correct**
normal_knee_2	Normal	0.178	0.345	0.114	0	0	0	**Correct**

C. OOD Analysis Discussion

The model achieves 4/6 correct predictions (66.7 % overall accuracy) on the OOD set, with the two failures concentrated in the ACL task. Key observations from Table IV and Fig. 6:

Normal knees: Both correctly predicted as negative across all three tasks, with low confidence scores (max P = 0.345), suggesting the model is well-calibrated for the healthy class even under domain shift.

Meniscal tears: Both correctly detected. meniscal_tear_2 shows notably high ACL confidence (P(ACL) = 0.843) despite the ground truth being meniscal-only, reflecting the model's tendency to predict co-morbid conditions when abnormality signals are strong.

ACL tears: The most challenging under domain shift. acl_tear_1 produces uniformly low probabilities (max = 0.268), likely because the image contrast and acquisition protocol differ substantially from the MRNet sagittal protocol, causing the backbone's feature representations to fall outside the training distribution manifold. These findings directly motivate the domain adaptation strategies outlined in Section VI.